

Assignment 3

Solve the following problems and upload the file of solutions in the D2L. You may type the solutions in Microsoft Word and upload the file, or you may scan your solutions by scanner and upload the file in the D2L (**due date: Thursday, November 9, 11:59 am**).

1.

- a. What is Primavera software?
- b. What are the advantages and disadvantages of this software compared to Microsoft Project?
- c. Which one (Primavera or Microsoft Project) is more expensive for the users?

(20 points)

2. The yearly demand for a seasonal, profitable item follows the distribution:

Demand (units)	Probability
1,000	.20
2,000	.30
3,000	.40
4,000	.10

A manufacturer is considering launching a project to produce this item and could produce it by one of three methods:

Method a. Use existing tools at a cost of \$6 per unit.

Method b. Buy cheap, special equipment for \$1,000. The value of the equipment at the end of the year (salvage value) is zero. The cost would be reduced to \$3 per unit.

Method c. Buy high-quality, special equipment for \$10,000 that can be depreciated over four years (one fourth of the cost each year). The cost with this equipment would be only \$2 per unit.

Set up this project as a decision tree to find whether the manufacturer should approve this project, and if so, which method of production to use to maximize profit.

Hint: Compare total annual costs. Assume that production must meet all demand; each unit demanded and sold means more profit.

(20 points)

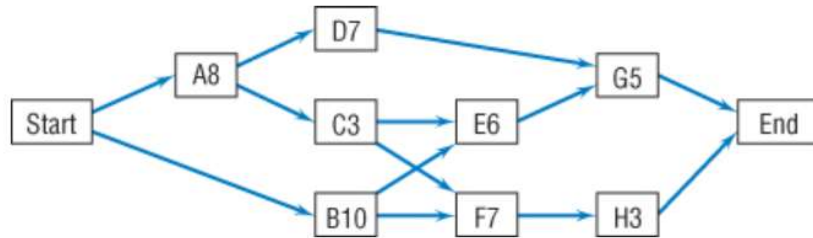
3.

- a. What is “slack” and why is it important in projects?
- b. How is the critical path determined?

(20 points)

4. (20 points)

Given the following network,



a. What is the critical path?

b. How long will it take to complete this project?

c. Can activity B be delayed without delaying the completion of the project? If so, how many days?

5. (20 points)

Given the estimated activity times below and the network in Exercise 4,

Activity	a	m	b
A	6.5	7.5	14.5
B	8.5	10.5	12.5
C	2.5	3.5	4.5
D	6.5	7.5	8.5
E	5.5	5.5	9.5
F	5.5	7.5	9.5
G	4.5	6.5	8.5
H	2.5	3.5	3.5

What is the probability that the project will be completed within (based on the path with the longest expected time):

a. 21 days?

b. 22 days?

c. 25 days?